

Numerical Verification and Proof-Guided Correction of a False Conjecture

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Numerical search has become a routine instrument of mathematical research: before a
2 conjecture is proved, it is tested against thousands of candidate counterexamples, and a
3 conjecture that survives is often written down as a result. We report a case in which this
4 practice failed, and in which the failure was found only because a human proof attempt
5 and a machine search corrected each other. In a theory of what it costs a decision-maker
6 to learn how its own actions change the data it later learns from, a lower bound says this
7 cost cannot fall below a fixed floor, and we asked whether an adversary who knows the
8 functional form of the response could beat it. An adversarial search over thousands of
9 experimental designs said no, with a best ratio of 1.005, and we recorded the floor as
10 structure-proof. The proof we then wrote established the bound only within two curvature
11 lengths of the operating point; the obstruction was a single inequality that fails outside
12 that region. Searching exactly where the proof stopped produced a counterexample: a
13 four-point design with one far probe of weight 3×10^{-4} , attaining ratio 0.8517, verified
14 at 50-digit precision. The original verdict was a search artifact; the corrected result is
15 two-sided and exact. We describe the loop that made the correction visible and place the
16 episode among eleven such corrections recorded in the same project.

17 1 Introduction

18 Machines now participate across the mathematical research loop: they suggest conjectures [Davies
19 et al., 2021], search for counterexamples [Wagner, 2021, Romera-Paredes et al., 2024], and prove
20 theorems [Trinh et al., 2024, Hubert et al., 2025]. Most reported successes run in one direction: the
21 machine proposes or refutes, and a person proves. The failure direction, in which a numerical verdict
22 is wrong and the record of how it went wrong is preserved, is rarely documented, though such records
23 are what calibrate trust in the next verdict.

24 This paper documents one such failure completely, inside a body of theory about the cost of self-
25 knowledge in performative prediction (Section 2). A machine search supported a claim. A human
26 proof attempt could establish the claim only on a region of the design space, and it named the
27 boundary of that region. A second machine search, restricted to the outside of that region, refuted
28 the claim. Neither half of the loop found the result alone. We then place it among ten more **pivots**,
29 measurement-forced changes to claims already written down, from the same project. This work is
30 not an autonomous mathematical agent: all proofs were written by people, and what was automated
31 is numerical checking, adversarial search over designs, and high-precision verification of witnesses
32 (Section 3).

33 **Contributions.** (1) A reproducible account of one proof-and-search correction cycle, from the
34 false belief through the failed proof to the frozen counterexample (Section 4, Theorem 2). (2) Ten
35 further pivots from the same project and an assessment of what the sample supports (Sections 4
36 and 5). (3) The infrastructure that made the correction detectable (a theorem register with per-result
37 proof status, preregistered checks, and continuous integration), shipped as an anonymized repository.
38 Related work is discussed in Appendix A.

2 Setting: the cost of self-knowledge

This section gives enough of the theory to see what the counterexample of Section 4 contradicts.

Consider a decision-maker whose deployed action changes the data it will later retrain on. In the REFLEX market of Nagarajan and Ashok [2026], which is the market used throughout this paper, a dealer who quotes a wide spread sees different order flow than one who quotes a tight spread, so the flow it must learn to price is partly its own doing. This is **performative prediction** [Perdomo et al., 2020, Hardt and Mendler-Düner, 2023]: retraining becomes a fixed-point iteration, and it converges to a point that is not the value-maximizing one. Algorithms that close this gap [Izzo et al., 2021, Miller et al., 2021, Drusvyatskiy and Xiao, 2023] need an estimate of how the world responds to the action, and they obtain it by *exploring*, that is, by deploying actions away from the current best guess and observing what changes. Those exploratory actions are real deployments, so they forgo value; the theory prices that loss.

Notation. Let h denote the scalar action, h^* the operating point the decision-maker would choose with full knowledge, and $d_t = h_t - h^*$ the deviation deployed at round t . Let ε be the *response slope*, the quantity to be estimated; σ^2 the observation noise variance; Φ the value function; and γ_{PO} its curvature at h^* , the rate at which value is lost away from h^* . The cost of exploration is the value forgone, $C_T = \sum_{t \leq T} \mathbb{E}[\Phi(h^*) - \Phi(h_t)]$. Labels such as T2 are the identifiers of the project’s theorem register; formal statements are in Appendix B.

Theorem 1 (Exchange rate; T2). *Along any stationary exploration policy, for the least-squares slope estimator, $\text{Var}(\hat{\varepsilon}) \times C_T = \frac{1}{2} \gamma_{PO} \sigma^2 (1 + o(1))$, independent of the exploration amplitude, shape, schedule, and of the contraction strength of the loop.*

In words, precision and cost trade off at a fixed exchange rate: exploring more reduces the variance of the estimate and raises the cost by exactly offsetting amounts, so their product is a constant. One chooses what to buy but not how much to pay. The reason is a cancellation: the information a design yields and the value it forgoes both grow with the same quantity, the spread of the deployed actions around h^* , so their ratio is fixed. Figure 1 shows the measured product lying on the predicted constant. Further results extend the floor to every policy and every estimator (T3, T4; Appendix B), and the theory was instantiated on the REFLEX corporate-bond calibration and on a second economy (Appendix D).

3 Methodology: the verification loop

Every claim in the project passes through the following procedure, listed in the order it occurred for the pivot of Section 4 (earlier pivots used subsets of the steps). (1) **Register the claim** with an identifier and a proof status (proved, proved for a stated class, conditional on a hypothesis, or numerically supported). (2) **Preregister a numerical check**: a closed form is compared to a simulation at a tolerance fixed before the run, and rows are reported as pass, fail, or drift. (3) **Search adversarially**: for a claim of the form “every design satisfies an inequality”, a random and locally refined search over designs looks for a violation. (4) **Attempt a proof**: if the proof succeeds only on a sub-region, its boundary is recorded and the search of step 3 is re-run restricted to the unreached region. (5) **Verify and record**: a candidate counterexample is frozen, recomputed with `mpmath` at 50-digit precision, and added to the suite that runs on every push; any change to a written claim is recorded, with its reason, in the code where it happened.

Steps 2, 3, the re-run in 4, and the verification in 5 are automated; every proof, every choice of what to conjecture, and the decision to search outside the proof’s reach were made by people. The machine never proposed a theorem; it refuted several.

4 Results: the structure-proofness pivot

The question. Theorem 1 concerns one estimator on one policy class, and the minimax floors T3 and T4 extend it to all policies and estimators. Both leave open the following. Suppose the adversary knows the functional form of the response, not only its slope: concretely, that it is $\tau_\theta(h) = C_0 + C_1 e^{-ch}$, an exponential with three unknown parameters, where c sets the curvature

88 and $1/c$ is the length over which the response bends. Knowing the form is real information, so it
 89 might permit an experimental design that beats the floor.

90 **The initial search verdict.** A design μ is a set of probe locations with weights, the fraction of
 91 the exploration budget spent at each. Its Fisher information $M(\mu)$ measures what the probes reveal
 92 about the three parameters, and the Cramér–Rao bound $g^\top M(\mu)^{-1}g$ is the smallest variance any
 93 unbiased estimator of the slope can achieve under μ , where g is the gradient of the slope with respect
 94 to the parameters. The relevant quantity is this bound times the design’s cost, normalized by the
 95 floor, $R(\mu) = (g^\top M(\mu)^{-1}g) \int (h - h^*)^2 d\mu$, so that the floor survives structural knowledge exactly
 96 when $R(\mu) \geq 1$ for every μ . Four thousand random designs plus 2,500 local refinements returned a
 97 minimum of $R = 1.0050$, and a scan over designs pairing two near-anchor points with one distant
 98 probe found distant probes monotonically worse, reaching $R = 40.9$ at the grid’s edge (Figure 2).
 99 Two independent strategies agreed, and we recorded the claim: the floor is structure-proof, with best
 100 attained ratio 1.005. This claim was later shown to be false.

101 **The proof attempt and its reach.** The proof attempt reduces $R(\mu)$ to a one-dimensional problem.
 102 It rewrites $R(\mu)$ as a ratio: the cost of the design, $\int d^2 d\mu$ with $d(h) = h - h^*$, divided by the smallest
 103 weighted size $\int \phi^2 d\mu$ of any function ϕ built from the model’s sensitivities and normalized to unit
 104 slope at the anchor (Appendix C). It therefore suffices to exhibit one such ϕ that is everywhere no
 105 larger than $|d|$; then the ratio is at least one. There is a canonical candidate, the unique such function
 106 whose value, slope, and curvature at the anchor match those of d :

$$\phi_0(h) = \frac{2}{c} - e^{-c(h-h^*)} \left(\frac{2}{c} + h - h^* \right).$$

107 Writing $x = h - h^*$, the required domination $|\phi_0| \leq |x|$ holds for every $x \geq -2/c$, with equality
 108 only at $x = 0$ and $x = -2/c$; this is an elementary argument (Appendix C). It is also the whole
 109 story: for $x < -2/c$ the domination is not merely unproved but false, and it fails exponentially. The
 110 proof therefore established the floor for every design supported on the reach $[h^* - 2/c, \infty)$, from
 111 two curvature lengths below the anchor upward, and it identified the boundary exactly.

112 **Targeted search and the witness.** The failed step is not a gap in the argument but a location.
 113 We re-ran the adversarial search twice. *Inside the reach*, a hardened search (three runs of 30,000
 114 random designs plus 10,000 refinements) pushed the minimum from 1.0050 down to $R = 1.00054$,
 115 consistent with the proof and with an infimum of exactly 1 that is approached but never attained.
 116 *Below the reach*, the same search found $R = 0.85164$. The witness is a four-point design with
 117 support $\{1.8726, 1.8665, 1.3073, 0.0500\}$ and weights $\{0.003329, 0.955702, 0.040697, 0.000272\}$,
 118 in the REFLEX market with $c = 1.5$ and anchor $h^* = 1.8461$. Every support point except the one
 119 at 0.05 lies inside the reach; the single below-reach probe of weight 3×10^{-4} is what breaks the
 120 floor. We froze the design and recomputed its ratio at 50-digit precision with `mpmath`, obtaining
 121 $0.851650\dots$; the check now runs in continuous integration on every push.

122 **Why the first search missed it.** First, the scan’s weight grid never went below 0.05, and the
 123 violating probe carries weight 3×10^{-4} . Second, the random search did generate distant probes, but
 124 always alongside a symmetric near-anchor pair, and symmetric-pair-plus-far-probe families never
 125 break the floor: their ratio approaches 1 from above as the probe weight vanishes, so the violation
 126 requires the asymmetric cluster. The earlier verdict was therefore neither noise nor a coding error but
 127 a correct summary of a search space that excluded the answer, and nothing inside that space could
 128 have revealed the exclusion.

129 **Theorem 2** (Structure-proofness, exact reach; T9). (i) Every estimable design supported in $[h^* -$
 130 $2/c, \infty)$ satisfies $R(\mu) > 1$ strictly, and $\inf_\mu R(\mu) = 1$, approached but not attained. Within reach,
 131 knowing the family cannot beat the floor. (ii) The reach is exact: the four-point witness above attains
 132 $R = 0.8517$.

133 **Interpretation.** Structure-proofness is an ignorance-of-curvature phenomenon with an exactly
 134 known extent: a design that reaches far enough below the anchor learns the curvature parameter c
 135 cheaply, and that saving is what the floor cannot account for. Two checks confirm this. With c known
 136 a priori, the floor breaks outright and keeps falling with no limit the scan reaches ($R = 0.3002$ at
 137 probe $t = 1.0$, 0.0425 at $t = 0.05$, 0.0348 and still decreasing at $t = 0.001$). And the reach is exactly

where the certified agent of Appendix D deploys: its trust region is $|d_t| \leq 0.8$, inside $2/c = 1.333$, so the constraint that keeps the agent safe is the one that keeps its lower bound valid. Scope: $R(\mu)$ and the floor are defined under the local-quadratic cost model (assumption A1), and under the true market cost the same witness has ratio 1.0112; the counterexample fixes the scope of the bound, not a way to beat the market.

Further pivots. One episode is an anecdote. Of the eleven pivots, three repeat the pattern above at smaller scale. **E1:** a saturation cap derived without noise failed by a factor of four against the noisy pipeline; the gap was a missing term, since retraining on noisy observations adds a stationary excitation floor of $(\sigma/\gamma)^2/(1-m^2)$ per step, itself now verified to within 0.2 to 6.4 percent (Figure 3). **E5:** an injected misspecification ae^{-2ch} had decayed to about 10^{-3} by the operating point, so the anchored fit correctly never detected it; replacing it with a linear leak the family cannot represent produced the crossover the theory predicts (Figure 4). **E7:** a claimed threefold advantage for shaped exploration collapsed to parity once the comparison arm was funded with the same energy (the unshaped arm had been given half the design’s energy per step); the corrected claim is sharper, since the identification cliff is a support-and-amplitude phenomenon, not a shape phenomenon. Each replacement was stronger than the claim it displaced.

5 Discussion and conclusion

What the sample supports. On this project, numerical falsification was cheap and frequent, and it improved claims rather than only deleting them. The sample supports no claim about proof automation, other areas of mathematics, or what happens without a person deciding where to look.

Infrastructure. A register lists all 26 results with proof status, so a conjecture cannot be cited as a theorem. A preregistered harness compares each measurement to a closed form at a tolerance fixed before the run; the current state is 36 rows with 0 failures and one deliberately out-of-scope cell reported as drift rather than excused. A derivation suite of 38 checks runs in continuous integration; its first runs falsified three drafted claims, one with the wrong sign. Every pivot is written into the code file where it occurred. We label the sharp constant $\frac{1}{2}$ in T4 as open and our own structure-proofness verdict as a search artifact: a record containing only successes cannot calibrate trust in the next verdict. This infrastructure is independent of the mathematics and is what we recommend other groups adopt.

6 Limitations

Theorem 1 is local: at finite exploration amplitude the identity is only approximate, and the deliberately out-of-scope saturating cell drifts by +16 percent. The value-budget floor T4 is proved only to within a factor of 13.5 of the conjectured constant. Which below-reach designs violate the floor of Theorem 2, and by how much at most, is open; and eleven pivots from one project by one group are a sample, not a study.

References

- Kenneth Appel and Wolfgang Haken. Every planar map is four colorable. Part I: Discharging. *Illinois Journal of Mathematics*, 21(3):429–490, 1977.
- Alexander Barzykin, Philippe Bergault, Olivier Guéant, and Malo Lemmel. Optimal quoting under adverse selection and price reading. *arXiv preprint arXiv:2508.20225*, 2026.
- Xavier Bombois, Gérard Scorletti, Michel Gevers, Paul M. J. Van den Hof, and Roland Hildebrand. Least costly identification experiment for control. *Automatica*, 42(10):1651–1662, 2006.
- Jonathan M. Borwein and David H. Bailey. *Mathematics by Experiment: Plausible Reasoning in the 21st Century*. A K Peters, 2nd edition, 2008.
- Daniele Bracale, Subha Maity, Yuekai Sun, and Moulinath Banerjee. Learning the distribution map in reverse causal performative prediction. In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 258 of *PMLR*, 2025.

- 184 Gavin Brown, Shlomi Hod, and Iden Kalemaj. Performative prediction in a stateful world. In
185 *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*,
186 2022.
- 187 Alex Davies, Petar Veličković, Lars Buesing, Sam Blackwell, Daniel Zheng, Nenad Tomašev, Richard
188 Tanburn, Peter Battaglia, Charles Blundell, András Juhász, Marc Lackenby, Geordie Williamson,
189 Demis Hassabis, and Pushmeet Kohli. Advancing mathematics by guiding human intuition with
190 AI. *Nature*, 600:70–74, 2021.
- 191 Arnoud V. den Boer. Dynamic pricing and learning: Historical origins, current research, and new
192 directions. *Surveys in Operations Research and Management Science*, 20(1):1–18, 2015.
- 193 Dmitriy Drusvyatskiy and Lin Xiao. Stochastic optimization with decision-dependent distributions.
194 *Mathematics of Operations Research*, 48(2):954–998, 2023.
- 195 A. A. Feldbaum. Dual control theory I. *Automation and Remote Control*, 21:874–880, 1960.
- 196 Richard D. Gill and Boris Y. Levit. Applications of the van Trees inequality: A Bayesian Cramér–Rao
197 bound. *Bernoulli*, 1(1–2):59–79, 1995.
- 198 Olivier Guéant, Charles-Albert Lehalle, and Joaquin Fernandez-Tapia. Dealing with the inventory
199 risk: A solution to the market making problem. *arXiv preprint arXiv:1105.3115*, 2013.
- 200 Thomas Hales, Mark Adams, Gertrud Bauer, Tat Dat Dang, John Harrison, Hoang Le Truong, Cezary
201 Kaliszyk, Victor Magron, Sean McLaughlin, Tat Thang Nguyen, et al. A formal proof of the
202 Kepler conjecture. *Forum of Mathematics, Pi*, 5, 2017.
- 203 Moritz Hardt and Celestine Mendler-Dünnér. Performative prediction: Past and future. *arXiv preprint*
204 *arXiv:2310.16608*, 2023.
- 205 Thomas Hubert, Rishi Mehta, Laurent Sartran, Miklós Z. Horváth, Goran Žužić, Eric Wieser, Aja
206 Huang, Julian Schrittwieser, and David Silver. Olympiad-level formal mathematical reasoning
207 with reinforcement learning. *Nature*, 651:607–613, 2025.
- 208 Zachary Izzo, Lexing Ying, and James Zou. How to learn when data reacts to your model: Performa-
209 tive gradient descent. In *Proceedings of the 38th International Conference on Machine Learning*
210 *(ICML)*, 2021.
- 211 Meena Jagadeesan, Tijana Zrnic, and Celestine Mendler-Dünnér. Regret minimization with performa-
212 tive feedback. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*,
213 2022.
- 214 N. Bora Keskin and Assaf Zeevi. Dynamic pricing with an unknown demand model: Asymptotically
215 optimal semi-myopic policies. *Operations Research*, 62(5):1142–1167, 2014.
- 216 Jack Kiefer and Jacob Wolfowitz. The equivalence of two extremum problems. *Canadian Journal of*
217 *Mathematics*, 12:363–366, 1960.
- 218 Tze Leung Lai and Herbert Robbins. Adaptive design and stochastic approximation. *The Annals of*
219 *Statistics*, 7(6):1196–1221, 1979.
- 220 Licong Lin and Tijana Zrnic. Plug-in performative optimization. In *Proceedings of the 41st*
221 *International Conference on Machine Learning (ICML)*, 2024.
- 222 John Miller, Juan C. Perdomo, and Tijana Zrnic. Outside the echo chamber: Optimizing the
223 performative risk. In *Proceedings of the 38th International Conference on Machine Learning*
224 *(ICML)*, 2021.
- 225 Vignesh Nagarajan and Shriraghav Ashok. REFLEX: Reflexive equilibrium fixed-point learning for
226 endogenous exchanges. *arXiv preprint arXiv:2608.16155*, 2026. URL <https://arxiv.org/abs/2608.16155>.
- 228 Juan C. Perdomo, Tijana Zrnic, Celestine Mendler-Dünnér, and Moritz Hardt. Performative prediction.
229 In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.

- 230 Friedrich Pukelsheim. *Optimal Design of Experiments*. SIAM Classics in Applied Mathematics,
231 2006.
- 232 Bernardino Romera-Paredes, Mohammadamin Barekatin, Alexander Novikov, Matej Balog,
233 M. Pawan Kumar, Emilien Dupont, Francisco J. R. Ruiz, Jordan S. Ellenberg, Pengming Wang,
234 Omar Fawzi, Pushmeet Kohli, and Alhussein Fawzi. Mathematical discoveries from program
235 search with large language models. *Nature*, 625:468–475, 2024.
- 236 Max Simchowitz and Dylan J. Foster. Naive exploration is optimal for online LQR. In *Proceedings*
237 *of the 37th International Conference on Machine Learning (ICML)*, 2020.
- 238 The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN*
239 *International Conference on Certified Programs and Proofs (CPP)*, pages 367–381, 2020.
- 240 Trieu H. Trinh, Yuhuai Wu, Quoc V. Le, He He, and Thang Luong. Solving olympiad geometry
241 without human demonstrations. *Nature*, 625:476–482, 2024.
- 242 Andrew Wagenmaker, Max Simchowitz, and Kevin Jamieson. Task-optimal exploration in linear
243 dynamical systems. In *Proceedings of the 38th International Conference on Machine Learning*
244 *(ICML)*, 2021.
- 245 Adam Zsolt Wagner. Constructions in combinatorics via neural networks. *arXiv preprint*
246 *arXiv:2104.14516*, 2021.

A Related work

Machines and people in mathematics. The closest precedent for the shape of our loop is the use of computation to guide intuition toward statements that people then prove [Davies et al., 2021]; there the direction is machine to human, whereas one leg of our eleventh pivot runs the other way, with the proof supplying a region rather than a candidate. A second line lets search refute conjectures directly, through reinforcement learning over constructions [Wagner, 2021] or program search guided by a language model [Romera-Paredes et al., 2024]; our search is far simpler than either, and it was thorough over a space that excluded the answer. A third line automates proof itself [Trinh et al., 2024, Hubert et al., 2025, The mathlib Community, 2020]; we do none of this, and our proofs are not formalized. Our harness sits instead in the tradition of machine-assisted proof [Appel and Haken, 1977, Hales et al., 2017] and experimental mathematics [Borwein and Bailey, 2008]: it is bookkeeping about how much a numerical verdict is worth.

The substrate. The theory builds on performative prediction [Perdomo et al., 2020, Brown et al., 2022, Jagadeesan et al., 2022, Bracale et al., 2025, Lin and Zrnic, 2024], learning-while-earning [den Boer, 2015, Keskin and Zeevi, 2014, Lai and Robbins, 1979], dual control and least-costly identification [Feldbaum, 1960, Bombois et al., 2006], van Trees bounds [Gill and Levit, 1995], classical and task-optimal design [Kiefer and Wolfowitz, 1960, Pukelsheim, 2006, Wagenmaker et al., 2021, Simchowitz and Foster, 2020], and the REFLEX market [Nagarajan and Ashok, 2026, Guéant et al., 2013, Barzykin et al., 2026].

B Assumptions and formal statements

This appendix collects the standing assumptions and the formal statements of the results the body argues from. Identifiers (L1, T2, T9, and so on) are the register identifiers used throughout the supplementary repository, so each statement can be matched to its derivation document, proof, and numerical checks. The register itself, with per-result proof status, is `mlxor-derivations/` `THEOREMS.md`.

B.1 Standing assumptions

A1 (local quadratic scope): Φ is C^3 and strongly concave on the operating interval, with expansions taken at h^* and third-order remainder bounded by $L_3 = \sup |\Phi'''|$. **A2** (exploration class): all policies are non-anticipating; A2-sym adds $\mathbb{E}[u_t | \mathcal{F}_t] = 0$, A2-adaptive is unrestricted, and costs are measured from the certainty-equivalent anchor. **A3** (noise exogeneity): the observation noise ζ_t is i.i.d., mean zero, variance σ^2 , independent of $\mathcal{F}_t$. **A3'** relaxes this to a feedback channel with gain ϕ . **A4** (trust region): $|d_t| \leq r$ pathwise. **A5** (stationary scale): $S_T = \Theta(T)$. **A6** (stable base loop): the retraining modulus satisfies $m < 1$.

The local dynamics are $d_{t+1} = -m d_t + u_t$ with $d_t = h_t - h^*$, observations $\tau_t = \tau(h^*) - \varepsilon d_t + \zeta_t$, and Φ satisfying $\Phi'(h^*) = \delta_0$ and $-\Phi'' = \gamma_{\text{PO}}$ at the operating point. S_{xx} denotes the centered design energy $\sum_{t \leq T} (d_t - \bar{d})^2$.

B.2 The exchange rate and its floors

Lemma 1 (Cost equivalence; L1). *Under A1–A3 with conditionally symmetric exploration, the expected incremental performative cost satisfies $C_T = \frac{1}{2} \gamma_{\text{PO}} \mathbb{E}[\sum_{t \leq T} d_t^2] + R_T$ with $|R_T| \leq \frac{L_3}{6} \mathbb{E} \sum_t |d_t|^3$, and the realized cost’s variance decomposes exactly as $\text{Var}(C_T^{\text{real}}) = \delta_0^2 \text{Var}(\sum_t d_t) + \frac{\gamma_{\text{PO}}^2}{4} \text{Var}(\sum_t d_t^2)$ with vanishing cross term.*

The body’s Theorem 1 is the asymptotic form of the following pathwise identity, which is what is proved in Appendix C.

Theorem 3 (The exchange rate, pathwise; T2). *Under A1, A2-sym, A3 and A6, in the stationary regime, on every realization,*

$$\text{Var}(\hat{\varepsilon} \mid \text{design}) \times \frac{1}{2} \gamma_{\text{PO}} \sum_{t \leq T} d_t^2 = \frac{1}{2} \gamma_{\text{PO}} \sigma^2 \left(1 + \frac{T \bar{d}^2}{S_{xx}} \right) = \frac{1}{2} \gamma_{\text{PO}} \sigma^2 (1 + O_P(1/T)),$$

independent of the exploration intensity, the horizon, and the modulus.

Theorem 4 (Minimax lower bound, deviation budget; T3). *Over all non-anticipating exploration policies with $\sum_{t \leq T} d_t^2 \leq D$ almost surely and all estimators, biased included, the minimax risk over an interval of width w satisfies $\inf \sup_{\varepsilon} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \sigma^2 / (D + \sigma^2(\pi/w)^2)$, so the exchange-rate product is a floor, attained with equality by symmetric probing with least squares. The argument is a van Trees reduction in the style of Gill and Levit [1995].*

Theorem 5 (Minimax lower bound, value budget; T4). *Under stationary-scale exploration and the certainty-equivalent anchor, for every non-anticipating policy and every estimator, $\sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \gamma_{\text{PO}} \sigma^2 / 27 (1 - O(T^{-1/2}))$, the supremum over the two-point response family at the minimax scale, the constant arising from a Le Cam reduction optimized at $c = 2/3$. The sharp constant $\frac{1}{2}$, against which no simulated policy fell by more than Monte Carlo error, is conjectured and is part of the open problem OPEN-1. It is not proved here or anywhere in the repository.*

B.3 Structure-proofness

Theorem 6 (Structure-proofness of the floor, exact reach; T9). *Fix the structural family $\tau_{\theta}(h) = C_0 + C_1 e^{-ch}$ ($C_1 > 0$, $c > 0$), the anchor $h^* > 0$, the sensitivity $s(h) = \partial_{\theta} \tau_{\theta}(h)$, and the estimand gradient $g = \partial_{\theta} \varepsilon(h^*) = -s'(h^*)$. For a finitely supported design μ define the normalized Cramér–Rao product*

$$R(\mu) = (g^{\top} M(\mu)^{-1} g) \int (h - h^*)^2 d\mu, \quad M(\mu) = \int s s^{\top} d\mu,$$

so that $R(\mu) \geq 1$ states that the family-knowing Cramér–Rao product $\text{Var}(\hat{\varepsilon}) \times C_T$ is at least the exchange-rate floor $\frac{1}{2} \gamma_{\text{PO}} \sigma^2$ under the local-quadratic (A1) cost, for unbiased estimators and, in the local-asymptotic-minimax sense, for all regular estimators. (i) Within reach the floor is structure-proof: if $\text{supp}(\mu) \subset [h^ - 2/c, \infty)$, then either $\varepsilon(h^*)$ is not estimable under μ , in which case the bound is infinite, or $R(\mu) \geq 1$, strictly whenever μ places mass off the two equality points $\{h^*, h^* - 2/c\}$, which is every estimable design, since a design supported on those two points alone leaves g outside the range of $M(\mu)$; and $\inf_{\mu} R(\mu) = 1$, approached but not attained by symmetric three-point designs collapsing at h^* . In particular the floor binds on every trust-region design with $r \leq 2/c$. (ii) The reach is exact: the pointwise domination underlying (i) fails strictly below $h^* - 2/c$, and an explicit four-point design with a single below-reach probe of weight 3×10^{-4} attains $R = 0.8517$, computer-verified at 50-digit precision as check V9.2.*

B.4 Other register results

The register also contains T1 (information saturation of the noiseless retraining trajectory), T5a and C5.1 (A-optimal exploration under a value budget and the price of isotropic jitter), T6 (Chebyshev unidentifiability of the curvature c from retraining trajectories), T7 (the anchoring crossover), L4 (perturbed modulus under an estimated response), and P7.1 (safety under pessimism given a confidence sequence). These are stated and proved in the supplementary repository (mlxor-derivations/latex/proofs.tex and the corresponding derivation documents) and are not restated here, since the body does not argue from them; they appear in the body only as the context that Theorems 1 and 2 sit in, and in Table 1 as verification rows.

C Proofs

This appendix proves the two results the body argues from: the exchange rate of Theorem 1, whose exact pathwise form is Theorem 3, and the two-sided structure-proofness result of Theorem 2, stated in full as Theorem 6. Both proofs are reproduced from the derivation repository without change; the check identifiers V2.1 and V9.1 to V9.4 refer to its verification suite.

C.1 The cost-equivalence lemma

Proof of Lemma L1 (cost equivalence). Taylor’s theorem with Lagrange remainder gives, for each t ,

$$\Phi(h^*) - \Phi(h_t) = -\delta_0 d_t + \frac{1}{2} \gamma_{\text{PO}} d_t^2 + r_t, \quad |r_t| \leq \frac{L_3}{6} |d_t|^3,$$

with $L_3 = \sup |\Phi'''|$ on the operating interval. Summing over $t \leq T$ and taking expectations, it remains to show $\mathbb{E}[\sum_t d_t] = 0$ under A2-sym. The recursion gives $\mathbb{E}[d_{t+1} | \mathcal{F}_t] = -m d_t + \mathbb{E}[u_t | \mathcal{F}_t] = -m d_t$, so $\mathbb{E}[d_{t+1}] = -m \mathbb{E}[d_t]$ by the tower property, and $\mathbb{E}[d_t] = (-m)^t \mathbb{E}[d_0] = 0$ when the loop starts at the operating point or in the stationary regime. The quadratic term contributes $\frac{1}{2} \gamma_{\text{PO}} \mathbb{E}[\sum_t d_t^2]$, which is the claim.

For the variance decomposition, write the realized cost (net of the cubic remainder) as $C = -\delta_0 X + \frac{1}{2} \gamma_{\text{PO}} Y$ with $X = \sum_t d_t$, $Y = \sum_t d_t^2$. Then $\text{Var}(C) = \delta_0^2 \text{Var}(X) + \frac{\gamma_{\text{PO}}^2}{4} \text{Var}(Y) - \delta_0 \gamma_{\text{PO}} \text{Cov}(X, Y)$. Under Gaussian innovations the vector $(d_1, \dots, d_T)$ is jointly Gaussian with mean zero, and every third moment of a zero-mean Gaussian vector vanishes; since $\text{Cov}(X, Y) = \sum_{t,s} \mathbb{E}[d_t d_s^2]$ is a sum of third moments, it is zero. (For non-Gaussian symmetric innovations the same conclusion follows from the sign symmetry $d \mapsto -d$ of the stationary law.) Verified: V0.1–V0.3, including the necessity check that an asymmetric rule breaks the expectation identity by exactly $-\delta_0 \mathbb{E}[\sum_t d_t]$. $\square$

C.2 The exchange rate

Proof of Theorem T2 (pathwise exchange rate). Under A3, conditionally on the design $\{d_t\}$ the observations are independent with means linear in ε and variance σ^2 , so the least-squares slope has conditional variance $\text{Var}(\hat{\varepsilon} | \text{design}) = \sigma^2 / S_{xx}$ (Gauss–Markov). Multiplying by the quadratic cost $\frac{1}{2} \gamma_{\text{PO}} \sum_t d_t^2$ and using $\sum_t d_t^2 = S_{xx} + T \bar{d}^2$ gives the displayed identity exactly, on every realization. In the stationary regime $\bar{d} = O_P(T^{-1/2})$ by the stationary long-run variance $\text{Var}(\sum_{t \leq T} d_t) = T v(1-m)/(1+m)(1+o(1))$, while $S_{xx}/T \rightarrow v$ a.s., so $T \bar{d}^2 / S_{xx} = O_P(1/T)$. Verified: V2.1 (machine precision). $\square$

C.3 Structure-proofness, and the exactness of the reach

Proof of Theorem T9 (structure-proofness within reach). Write $x = h - h^*$, $d(h) = x$, and $V = \text{span}\{1, e^{-ch}, h e^{-ch}\}$, the span of the components of s (a bijective correspondence $u \leftrightarrow \phi_u = u^\top s$ between $\mathbb{R}^3$ and V , since $C_1 \neq 0$). Differentiating s gives $s'(h) = (0, -c e^{-ch}, C_1 e^{-ch}(ch - 1))^\top$, so $g = -s'(h^*)$ componentwise, which is the stated estimand gradient.

Step 1: one-dimensional reduction. For $M(\mu) \succ 0$ the generalized Rayleigh identity gives

$$g^\top M(\mu)^{-1} g = \sup_{u \neq 0} \frac{(u^\top g)^2}{u^\top M(\mu) u} = \sup_{\phi \in V} \frac{\phi'(h^*)^2}{\int \phi^2 d\mu} = \left(\inf \left\{ \int \phi^2 d\mu : \phi \in V, \phi'(h^*) = 1 \right\} \right)^{-1},$$

using $u^\top M u = \int \phi_u^2 d\mu$ and $u^\top g = -\phi'_u(h^*)$ (the sign is lost in the square). Hence

$$R(\mu) = \frac{\int d^2 d\mu}{\inf \{ \int \phi^2 d\mu : \phi \in V, \phi'(h^*) = 1 \}}.$$

Step 2: the candidate and its pointwise domination. Let

$$\phi_0(h) = \frac{2}{c} - e^{-cx} \left(\frac{2}{c} + x \right) = \frac{2}{c} \cdot 1 - e^{ch^*} \left(\frac{2}{c} - h^* \right) e^{-ch} - e^{ch^*} h e^{-ch} \in V,$$

the unique element of V with 2-jet $(0, 1, 0)$ at h^* (the 2-jet map is invertible on V : in the basis $\{1, e^{-ch}, h e^{-ch}\}$ its determinant is $c^2 e^{-2ch^*} \neq 0$); the expansion is $\phi_0(x) = x - (c^2/6)x^3 + O(x^4)$. Writing $f(x) = \frac{2}{c} - e^{-cx}(\frac{2}{c} + x)$ we claim

$$|f(x)| \leq |x| \text{ for all } x \geq -2/c, \text{ with equality iff } x \in \{0, -2/c\},$$

and $|f(x)| > |x|$ strictly for every $x < -2/c$. Indeed: (a) $v(x) = f(x) - x$ has $v(0) = 0$ and $v'(x) = e^{-cx}(1 + cx) - 1 < 0$ for $x \neq 0$ (since $1 + t < e^t$ for $t = cx \neq 0$); $v' \leq 0$ with equality only at the isolated point 0 makes v strictly decreasing on $\mathbb{R}$, so $f > x$ on $x < 0$ and $f < x$ on $x > 0$. (b) $f(0) = 0$ and $f'(x) = e^{-cx}(1 + cx) > 0$ for $x \geq 0$, so $f \geq 0$ on $[0, \infty)$ and, with (a), $0 \leq f \leq x$ there. (c) $q(x) = f(x) + x$ has $q(-2/c) = 0 = q(0)$ and $q'(x) = e^{-cx}(1 + cx) + 1$, where $r(x) = e^{-cx}(1 + cx)$ is strictly increasing on $x < 0$ ($r'(x) = -c^2 x e^{-cx} > 0$) from $r(-2/c) = -e^2 < -1$ to $r(0) = 1$; so q' changes sign exactly once on $(-2/c, 0)$, from negative

375 to positive, and $q \leq 0$ there with equality only at the endpoints; with (a) ($f \geq x$, equality only
376 at 0) this gives $x \leq f \leq -x$ on $[-2/c, 0]$. (d) For $x < -2/c$, $r(x) < -e^2$ gives $q' < 0$, so
377 moving left from $-2/c$ the function q increases above 0: $f(x) > -x = |x| > 0$, and in fact
378 $f(x) = 2/c + e^{c|x|}(|x| - 2/c)$ grows exponentially.

379 *Step 3: the bound.* If $\text{supp}(\mu) \subset [h^* - 2/c, \infty)$, Step 2 gives $\int \phi_0^2 d\mu \leq \int d^2 d\mu$, so by Step 1,
380 $R(\mu) \geq \int d^2 d\mu / \int \phi_0^2 d\mu \geq 1$. Strictness: equality in Step 2 holds only at $x \in \{0, -2/c\}$, so any
381 μ placing positive mass off $\{h^*, h^* - 2/c\}$ (in particular any with three distinct support points)
382 has $\int \phi_0^2 d\mu < \int d^2 d\mu$ and $R(\mu) > 1$. If $M(\mu)$ is singular, two cases: $g \notin \text{range } M(\mu)$ makes
383 $\varepsilon(h^*)$ non-estimable (infinite bound); $g \in \text{range } M(\mu)$, which does occur for special two-point
384 designs (a one-parameter family of in-reach pairs; their finite pseudo-inverse products were checked
385 to obey the bound, e.g. $R = 1.063$ at the pair $\{1.627, 2.127\}$), keeps the sup-representation valid
386 over $\{u : u^\top M u > 0\}$, and $\int \phi_0^2 d\mu = 0$ would put $u_0 \in \text{null } M$ with $u_0^\top g = -1 \neq 0$, contradicting
387 $g \in \text{range } M$; so $\int \phi_0^2 d\mu > 0$ and the bound goes through verbatim. Throughout, the Cramér–Rao
388 functional bounds unbiased estimation, and all regular estimators in the local-asymptotic-minimax
389 sense.

390 *Step 4: the infimum is 1.* The Cramér–Rao functional is invariant under C^1 reparametrization,
391 and the 2-jet map $\theta \mapsto (\tau_\theta(h^*), \varepsilon(h^*), \tau_\theta''(h^*))$ is a local diffeomorphism: its Jacobian $[s(h^*) -$
392 $s'(h^*) \ s''(h^*)]$ has determinant $C_1 c^2 e^{-2ch^*} \neq 0$ (direct computation; V9.4). In jet coordinates η the
393 model reads $\tau = \eta_1 - \eta_2 x + \frac{1}{2} \eta_3 x^2 + O(x^3)$ with sensitivity $\tilde{s}(x) = (1, -x, x^2/2)^\top + O(x^3)$, and
394 the estimand is η_2 . For the symmetric three-point design μ_δ uniform on $\{h^* - \delta, h^*, h^* + \delta\}$, scale
395 by $D = \text{diag}(1, \delta, \delta^2)$: $M_\eta(\mu_\delta) = D \hat{H}_\delta D$ with $\hat{H}_\delta \rightarrow \hat{H}_0$ invertible (the exact quadratic-model
396 moment matrix), so $(M_\eta^{-1})_{22} = \delta^{-2} (\hat{H}_\delta^{-1})_{22}$ and $R(\mu_\delta) \rightarrow R_{\text{quad}}$. In the exact quadratic model,
397 for any symmetric design (odd moments $m_1 = m_3 = 0$), the $(2, 2)$ cofactor gives $(M_\eta^{-1})_{22} = 1/m_2$
398 and $R_{\text{quad}} = m_2 \cdot (1/m_2) = 1$. Hence $R(\mu_\delta) \rightarrow 1$ (measured rate $R(\mu_\delta) - 1 = O(\delta^2)$; V9.2), the
399 infimum is 1, and by Step 3 it is not attained.

400 *Step 5: the reach is exact.* Below $h^* - 2/c$ the domination of Step 2 fails strictly, and the fail-
401 ure is realized by a design, not only by the candidate: the frozen four-point witness (support
402 $\{1.8726, 1.8665, 1.3073, 0.0500\}$, weights $\{0.003329, 0.955702, 0.040697, 0.000272\}$ after normal-
403 ization, in the register market $c = 1.5$, frozen anchor $h^* = 1.8461$) has $R = 0.851650\dots$, verified at
404 50-digit precision (mpmath) and reproduced in float64 by `run_open1.py` (V9.2). All its support
405 except the probe at 0.05 lies within reach: one vanishing-weight below-reach probe breaks the floor.
406 Symmetric pair-plus-far-probe families do not break it (their ratio approaches 1 from above as the
407 probe weight vanishes); the violation requires the asymmetric cluster, which is why coarse searches
408 missed it. Verified: V9.1–V9.4. $\square$

409 D Verification rows

410 Table 1 lists selected rows of the preregistered harness. Appendix E plots the rows the body refers to.

Table 1: Selected rows from the preregistered check harness. The full table (36 rows, 0 FAIL), all seeds, tolerances, and code are in the supplementary repository. The drift row is a deliberately out-of-scope cell, measured and reported rather than excused.

Claim (register)	Measured	Predicted	Verdict
Exchange rate, $m = 0.3$, two amplitudes (T2)	0.01062, 0.01070	0.01056	pass
Exchange rate, $m = 0.6$, two amplitudes (T2)	0.00549, 0.00569	0.00535	pass
Saturating environment (outside A1)	+16% over rate	out of scope	drift
Feedback floor, $m = 0.3/0.6/0.85$ (T1 corollary)	within 0.2–6.4%	$(\sigma/\gamma)^2/(1 - m^2)$	pass
Design arm $\text{sd}(\hat{c})$; jitter/design ratio (T6)	0.234; $7.8\times$	Fisher 0.248; ≥ 3	pass
Certified agent reaches h_{PO} ; trust region (L4)	1.258; 0 violations	1.296; 0	pass
Isotropic/A-optimal risk ratio, $d = 8$ (T5a)	1.529	$F = 1.506$	pass
Nonparametric MSE at (B, T) optimum (T7)	0.0132	0.0123	pass
Second economy, 4 cells (T2, exact in LQ); agent	0.384, 0.192; 12.65	0.384, 0.192; 12.5	pass
Multi-book shaped/isotropic steady-state risk (T5a)	1.20 (flat control 1.00)	$> 1.15 (\approx 1)$	pass
Real-data calibration port, 10 cells	10/10 reproduce	within 0.8%	pass
Within-reach min CR ratio; witness beyond (T9)	1.0005; 0.8517	≥ 1 ; < 1	pass

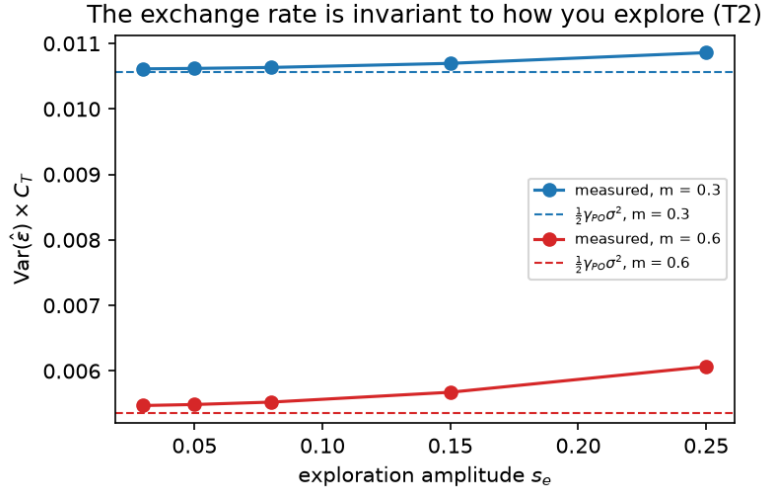


Figure 1: The exchange rate is invariant to how one explores (T2). Measured $\text{Var}(\hat{\epsilon}) \times C_T$ against exploration amplitude at two contraction moduli m ; the dashed lines are the predicted constant $\frac{1}{2}\gamma_{PO}\sigma^2$. The product does not depend on how much one explores, only on the curvature and the noise.

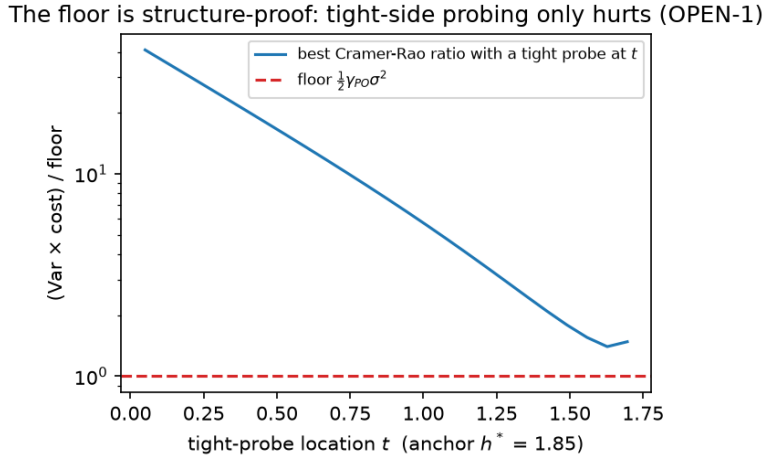


Figure 2: The search that produced the false verdict (OPEN-1). Best Cramér–Rao ratio with one tight probe at location t , over the coarse weight grid with $w \geq 0.05$; the floor is the dashed line. On this family, tight probes are monotonically worse the further they sit from the anchor, which is what we believed in general. The counterexample of Section 4 lies outside this family: its probe weight is 3×10^{-4} , inside an asymmetric cluster the search never generated.

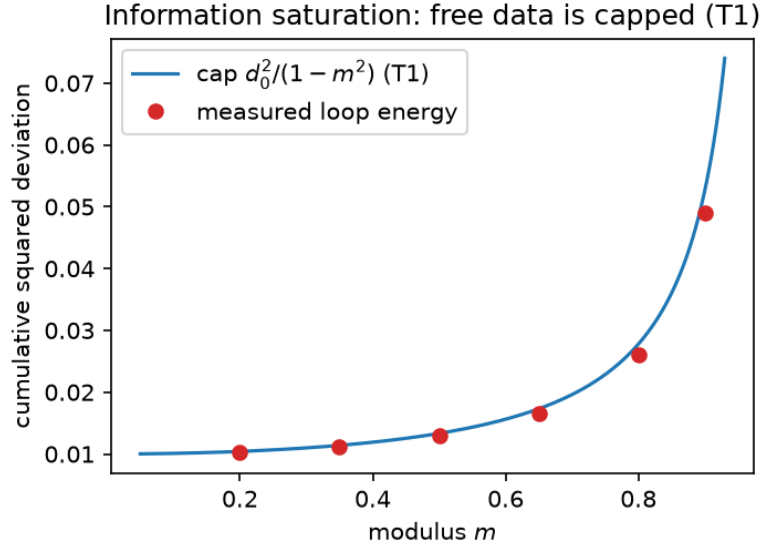


Figure 3: Information saturation (T1). Cumulative squared deviation of the noiseless retraining trajectory against the modulus m ; the cap $d_0^2/(1 - m^2)$ (line) matches the measured loop energy (points). Without deliberate exploration, the information a retraining loop reveals about its own performativity is bounded, and the bound grows with m . Pivot E1 added to this picture the excitation floor $(\sigma/\gamma)^2/(1 - m^2)$ that noisy observations contribute (Table 1, feedback-floor row).

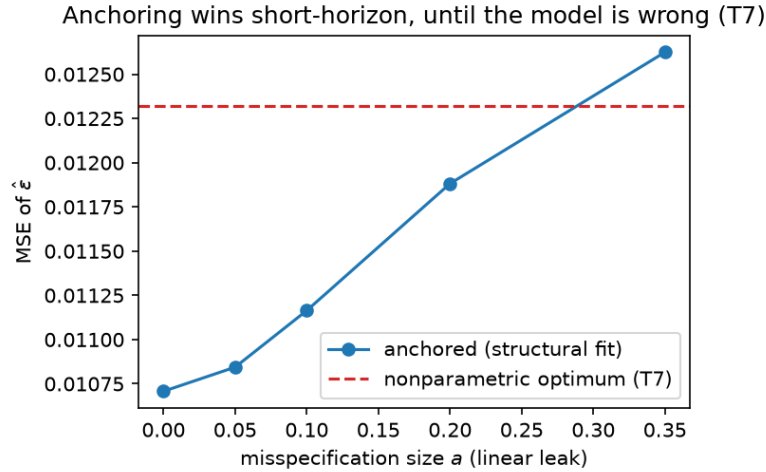


Figure 4: Anchoring crossover (T7). Mean squared error of the anchored structural fit against the size a of a linear misspecification the family cannot represent, compared with the budget-and-horizon-optimal nonparametric estimator (dashed). Anchoring wins at small misspecification and loses past the crossover near $a \approx 0.29$, as T7 predicts. Pivot E5 replaced an exponential misspecification, which had decayed to about 10^{-3} by the operating point and was correctly invisible to the fit, with this linear leak.

Certification before exploitation: the gate timeline (shaded = frozen)

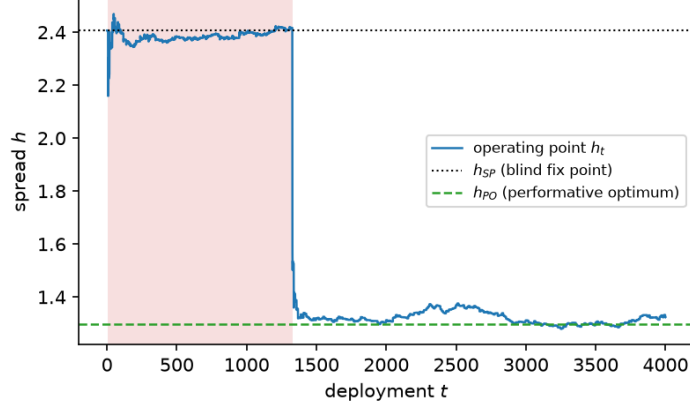


Figure 5: Certification before exploitation (L4, P7.1). Operating point h_t of the certified agent over 4,000 deployments. In the shaded interval the confidence sequence cannot yet certify that correcting is safe, so the agent stays frozen near the blind fixed point h_{SP} ; once certified, it moves to the performative optimum h_{PO} and remains inside the trust region (Table 1: 0 violations).

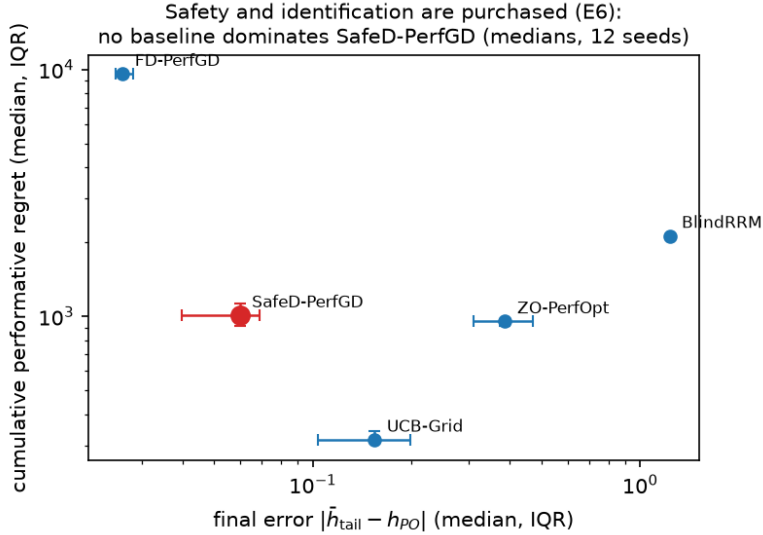


Figure 6: Baseline comparison (E6). Median final error $|\bar{h}_{tail} - h_{PO}|$ against median cumulative performative regret over 12 seeds (bars: interquartile range) for the certified agent (SafeD-PerfGD) and four baselines. No baseline dominates it: FD-PerfGD attains lower error at roughly ten times the regret, and UCB-Grid lower regret at higher error. Safety and identification are purchased, not free.

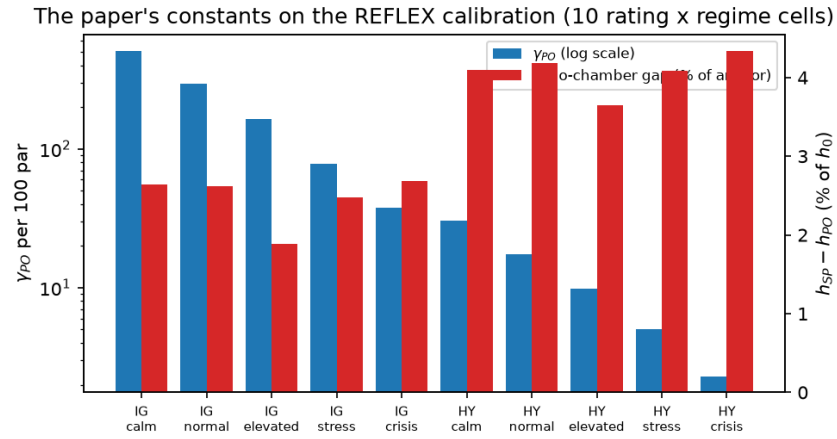


Figure 7: Real-data calibration (Table 1, calibration row). The curvature γ_{PO} (log scale, left axis) and the echo-chamber gap $h_{SP} - h_{PO}$ as a percentage of the anchor (right axis) across the ten rating-by-regime cells of the REFLEX corporate-bond calibration [Nagarajan and Ashok, 2026] (IG: investment grade; HY: high yield). All ten cells reproduce within 0.8 percent.

F Reproducibility

What ships. The supplementary repository contains the derivation documents and the theorem register (26 results, 38 numerical derivation checks), the simulation pipeline (environments, estimators, designs, agents, baselines; 36 measured-versus-predicted rows; 9 unit tests), the real-data port with its 10-cell validation, and the eleven-pivot record in the pipeline README. Both verification suites run in continuous integration on every push in the project repository; the two workflow files are not part of this anonymized copy. The structural market used throughout, and the corporate-bond calibration behind the real-data row of Table 1, are those of REFLEX [Nagarajan and Ashok, 2026], on which this work builds. All results are deterministic from seeds, `numpy`-only, and CPU-only. Reported final errors are 300-step tail averages over the deployed path, stable within 0.01 under horizon perturbation, with a worst measured spread of 0.0062.

What is stated here but not proved here. This version proves Theorems 1 and 2, the two results the body argues from. The minimax floors and the design and certification results (T3, T4, T5a, C5.1, T6, T7, L4, P7.1) are proved in the supplementary repository, in `mlxor-derivations/latex/proofs.tex` and the corresponding derivation documents, with the register recording each result’s proof status.

Provenance of the numbers. Every quantity in the body and in Table 1 is copied from a committed artifact: `results/RESULTS.md` for the harness rows, `results/OPEN1.md` for the reach, the witness, and the search history, `results/REALDATA.md` for the calibration port, `results/STABILITY.md` for the tail-average stability, `results/full_run.log` for quantities not surfaced in the summary tables, and `THEOREMS.md` for the register. The runs behind them were executed and recorded on 1 September 2026. This version re-narrates them and does not re-run them, since none of its changes touch their inputs.